

CubeSats and orbital debris

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1 Introduction

Humans have been sending objects into space since 1957 with the launch of Sputnik. Since all operators of spacecraft share the same space, the US Department of Defense created a central cataloging system called the US Space Surveillance Network (USSSN). The USSSN catalogs objects in low Earth orbit (LEO) >10 cm in diameter and >1 m in diameter in geosynchronous orbit (GEO). As of June 2019 more than 23,000 pieces of orbital debris appear in the catalog, and this number is growing [1]. Of these 23,000 pieces, only about 5% of these are active spacecraft. Scientists believe that $\sim 500,000$ objects range in diameter of 1–10 cm, most of which are inactive. They believe the total number of objects larger than 1 mm in diameter is >100 million, which includes both cataloged and uncataloged pieces. Due to the increased number of launches each year, the number of payloads housed on a launch, and an increase in fragmentation events, the rise in the number of objects in the catalog has been staggering. Fig. 1 shows the particle environment as seen from outside of GEO with each dot representing an object in the catalog [2]. The dots are not to scale for the size of the particle.

Orbital debris, also known as space debris, is defined as any human-made object in orbit about the Earth no longer serving its useful and/or designed purpose. The Inter-agency Space Debris Coordination Committee (IADC) is an international organization that looks to coordinate efforts dealing with orbital debris. A steering committee exists with four different working groups: measurements, environment and database, protection, and mitigation. This group reports to the United Nations Committee on the Peaceful Uses of Outer Space (UN-COPUOS). One example of the group's efforts is the mitigation working group that determines guidelines for spacecraft operators to follow for mitigation of debris and for limiting debris creation. This chapter introduces the reader to human-made orbital debris and explores areas in which the CubeSat community can be a part of the orbital debris solution.

1.1 Orbital debris facts

There are many sources of orbital debris: derelict satellites, upper stages of launch vehicles, debris released during operations, debris created by explosions or collisions (intentional or unintended), solid rocket motor slag, and even paint released during thermal

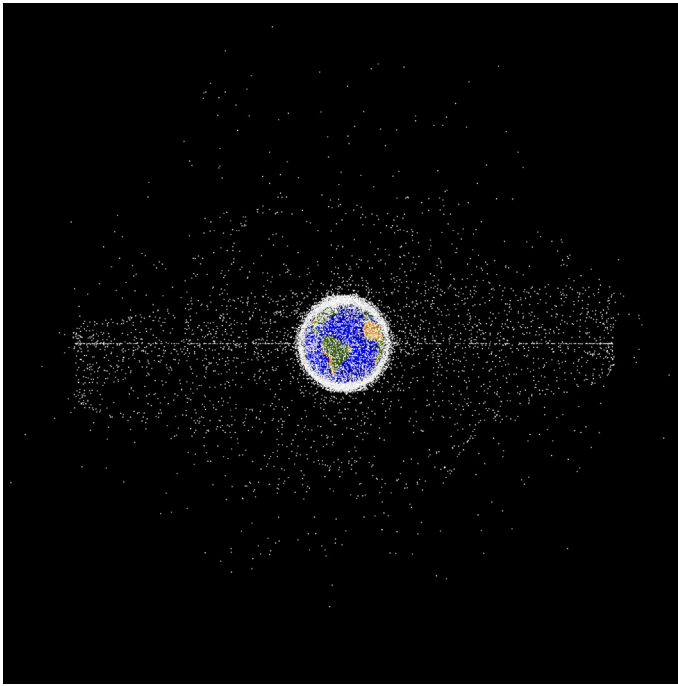


Fig. 1 Cataloged objects (shown as *dots*) around the Earth.

stress or from small particle impacts. [Fig. 2](#) shows the breakdown of the 23,000 cataloged objects in space [1]. As shown in the figure, fragmentation debris is the largest contributor to the orbital debris environment. Fragmentation debris stems from object explosion or collisions. Prior to 2007 the principal debris source was old rocket body upper stages. However, the intentional collision in 2007 of the Chinese weather satellite Fengyun-1C and the unintentional collision of an American satellite (Iridium 33) and a Russian satellite (Kosmos 2251) in 2009 drastically increased the number of fragmentation objects, accounting for nearly one-third of all cataloged debris.

Because debris travels at the same rate as objects in space, around 7 km/s in LEO, the average impact speed in LEO is close to 10 km/s. [Fig. 3](#) demonstrates the damage caused by a small particle that created a 2 mm in diameter crater in the window of the space shuttle [2]. Collisions with even a small particle will create damage to the spacecraft. Approximately 3–4 times per year the International Space Station maneuvers away from a possible collision with a larger cataloged object. However, objects with no propulsion system have no way of making such maneuvers. In addition, fragmentations usually create many smaller pieces than their original parent objects, which increase the number of smaller objects quickly. Since objects smaller than 10 cm in diameter cannot be tracked reliably, even objects with propulsion cannot always evade collision. While some satellites carry shielding against debris, such shielding creates extra mass and costs money, so shielding is often not an option. Therefore the most important action scientists can take against the growth of orbital debris is to prevent creating more of it, as much as possible.

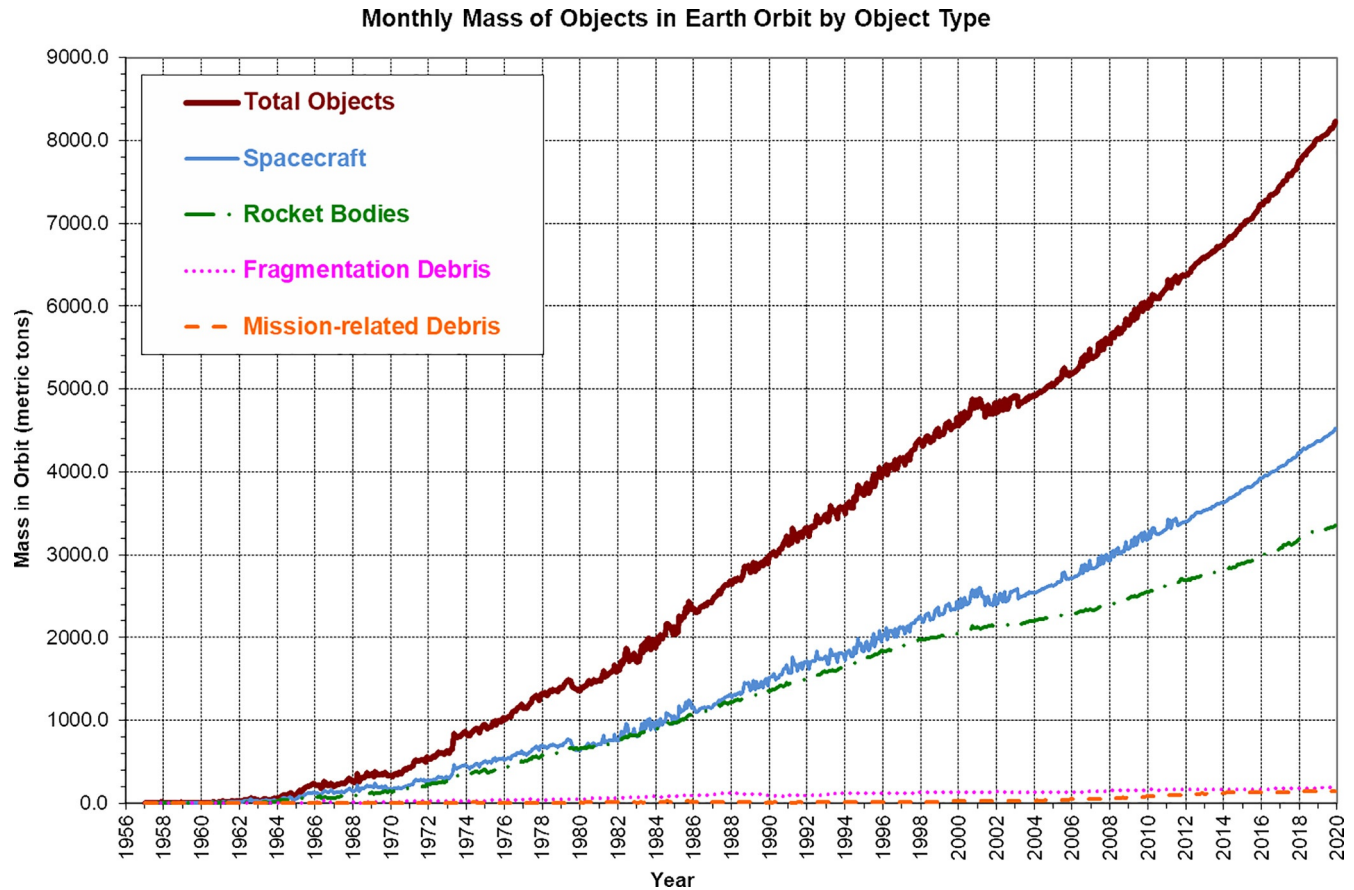


Fig. 2 Number of cataloged Earth-orbiting objects over time.

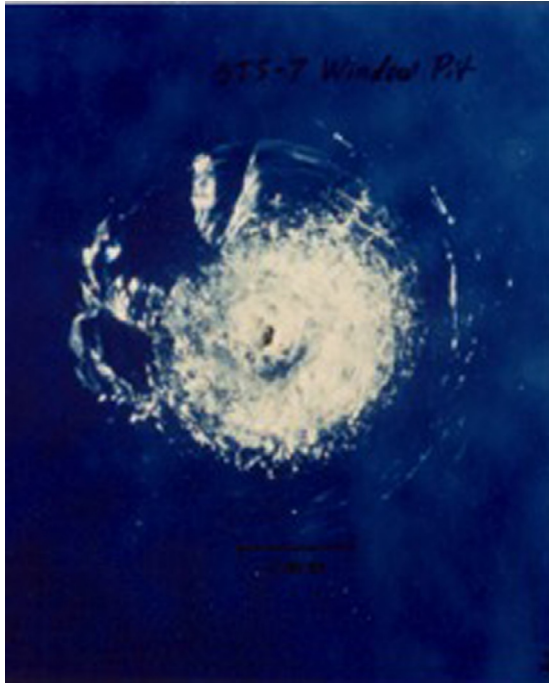


Fig. 3 Impact seen on an STS-7 window.

1.2 Use of reentry for mitigation

One method operators use to mitigate the growth of the orbital debris population is to allow the object to reenter the Earth's atmosphere, burning up in the atmosphere at the end of the useful mission. Objects in low Earth orbit (LEO) (~ 2000 km altitude and lower) are subject to drag from the atmosphere, and most objects will eventually reenter naturally. The time it takes an object to reenter is a function of its size, mass, altitude, and attitude. Further discussion of this method is presented in [Section 2.1](#). However, this mitigation method becomes more effective when satellite operators deorbit their own satellite and rocket body at the end of the mission.

1.3 CubeSats

CubeSats often get into orbit as a secondary payload. In most cases, this means they have less control over the orbit altitude into which they are placed. Due to the variations in CubeSat missions and capabilities, the following sections discuss ways in which CubeSats can meet the orbital debris requirements and potentially be used as a platform to study or remove orbital debris.

2 CubeSat mission analysis

2.1 Determining a CubeSat's orbit lifetime

A key aspect of a CubeSat mission is the amount of time it spends on orbit; this directly affects the likelihood it will collide with micrometeoroids or orbital debris, potentially making the on-orbit environment worse for future operations, because the longer the on-orbit time, the more the likelihood of a collision. The on-orbit lifetime of a spacecraft is highly dependent on the average altitude of the initial orbit: CubeSats deployed from the International Space Station (~400 km initial altitude) will typically remain on orbit fewer than 3 years, while those deployed in a Sun-synchronous orbit at 700-km altitude may still be on orbit more than 75 years later. These longer times can be contrasted with the typical operational lifetimes of CubeSats of fewer than 3 years, determined by the hardware reliability, the effects of radiation and thermal cycles, and other random failures.

Clearly, even though CubeSats are small in both mass and size (normally 1 kg and 1000 cm³), they can produce an outsized effect in the amount of debris in orbit when they are placed in orbits where their lifetimes are longer than expected. The expected lifetime of a satellite can be predicted by computing an area-to-mass ratio (or ballistic coefficient) and using a standard orbit propagator, such as SGP, and propagating forward in time until the perigee altitude drops below 50 km.

A rough calculation of orbital lifetime can be obtained using a high-order Runge-Kutta ordinary differential equation (ODE) solver, like ODE45 in MATLAB. Using initial position and velocity vectors, one can predict the altitude of the object over time in the presence of atmospheric drag by using the equation of motion shown in Eq. (1) and integrating it:

$$a = -\frac{\mu * \mathbf{r}}{r^3} - \frac{1}{2} \rho C_d \frac{A}{m} v_{rel}^2 \frac{\mathbf{v}_{rel}}{v_{rel}}, \quad (1)$$

where a is the acceleration (that is integrated to get velocity), μ is the gravitational constant (398,600 km³/s² for the Earth), r is the magnitude of the position vector, $\mathbf{r}$ is the position vector, ρ is the density of the atmosphere that is a function of altitude, C_d is the coefficient of drag (usually assumed to be 2.1), A is the area in the velocity vector direction, m is the spacecraft mass, v_{rel} is the speed of the spacecraft relative to the rotating atmosphere, and $\mathbf{v}_{rel}$ is the velocity vector of the spacecraft. The first half of the formula is the acceleration due to the central body, and the second half is the acceleration due to drag. One can see the dependence the area-to-mass (A/m) ratio has on the orbital lifetime of an object. An object with a larger area and a smaller mass, like the case of a solar sail, will greatly reduce the orbital lifetime of an object. With knowledge of the CubeSat's area, mass, position, and velocity, the integration of Eq. (1) over time will give the new position and velocity, and one can determine the estimated orbital lifetime.

2.2 *Applicable orbital debris requirements*

CubeSats that launch with either the NASA Launch Service Program (LSP) or US Air Force or wish to broadcast from orbit to a station in the United States are subject to certain requirements (set forth in Refs. [3–5]). At the international level, similar guidelines were developed by the Interagency Space Debris Coordination Committee (IADC) in 2002 and revised in 2007 [6]. These requirements include quantitative requirements for lifetime and quantity of debris released during mission operations, the likelihood of collision with large objects on orbit, the reliability of end-of-mission maneuvers (if any are planned), and for the expectation of casualty to the human ground population if an atmospheric reentry is planned. Most CubeSat missions will not have pressurized volumes or thrusters onboard, so a large fraction of the orbital debris mitigation requirements is not applicable. In addition, the CubeSat design specifications (1U and 6U; see Refs. [7, 8]) state that if materials other than aluminum are used, “developers should contact the Mission Integrator or dispenser manufacturer.” Stainless steel, titanium, and other materials with high melting point temperatures are likely to survive atmospheric reentry to ground impact, increasing the probability of ground casualty, and so should be avoided if possible.

The final orbit debris mitigation requirements of interest are the maximum orbit lifetime (typically limited to 25 years after end of mission per the guidelines from the IADC) and the large-object collision probability. Satisfying these requirements is strongly dependent on the area-to-mass ratio of the CubeSat; a 3U CubeSat with a mass of 5 kg has an area-to-mass ratio of $\sim 0.007 \text{ m}^2/\text{kg}$, which corresponds to an on-orbit lifetime of 80 years when deployed in a 700-km circular orbit at 98.4-degree inclination. A general rule is that an object without propulsion needs to be no higher than 600–700 km in altitude to abide with the 25-year guideline. Since the cross-sectional area of the CubeSat is so small, the probability that any CubeSat will collide with another tracked object during its orbital lifetime is $< 10^{-6}$; however, any decrease in orbit lifetime will reduce the potential effect of the satellite on the future growth of the orbital debris environment.

Operators can reduce the on-orbit lifetime of their satellite in four main ways (summarized in Table 1): increasing area-to-mass ratio by deploying a drag sail, using small chemical thrusters, using electrical propulsion such as ion or plasma thrusters, and by deploying a tether (passive or electrodynamic). Drag sails are perhaps the simplest to understand—these are typically made of a thin film of metal or polymer and

Table 1 Summary of orbit lifetime reduction methods.

Lifetime reduction method	Residual lifetime	Mitigate casualty risk?
Drag sail	Months to years	No
Chemical thruster	Minutes to years	Only if direct reentry
Electric propulsion	Days to years	No
Passive tether	Days to years	No
Electrodynamic tether	Days to years	No

can increase the cross-sectional area of a satellite by multiple orders of magnitude (and thus reduce the orbit lifetime dramatically). Chemical thrusters can provide enough thrust to perform a controlled reentry (i.e., reduce the perigee altitude to below 50 km) but require a large portion of the mass and volume of the satellite, and many launch providers will not accept CubeSats with chemical propellants onboard. Electrical propulsion, such as ion and plasma thrusters, performs much the same duties as chemical thrusters but has much lower thrust and operates over a much longer period of time than chemical thrusters. These do not typically have the capability of performing a controlled reentry, but can lower the orbit altitude to reduce orbit lifetime, using much less propellant (at the expense of high electrical power requirements). Tethers are simply long cords, made of polymers or conductive materials, depending on the application. Polymer tethers are used similarly to drag sails to increase the area exposed to atmospheric drag. Conductive tethers run a current down the cord and induce a Lorentz force using the Earth's magnetic field. All four of these methods are technically feasible, but each has significant drawbacks: Drag sails and tethers require a deployment at the end of mission, and thus may have reliability concerns, while chemical and electric propulsion may have additional launch safety concerns and restrictions.

If an object cannot reenter within the time, a graveyard orbit is recommended by the IADC. The final altitude must be above 2000 km and below 35,586 km; however, the commonly used 12-h orbit (GPS) altitude should be avoided. Due to the delta- v needed to achieve this requirement, only satellites above 1400-km altitude initially can obtain the desired altitude for graveyard in LEO. CubeSats in GEO will not be able to reenter and thus need to follow different guidelines for end-of-life (EOL) than the LEO satellites. The operators will need to increase the perigee altitude of the spacecraft to $235 \text{ km} + (C_r * A/m)$, where the C_r is the solar radiation pressure coefficient and A/m is the area related to the mass of the spacecraft. In addition, the eccentricity must be ≤ 0.003 .

3 CubeSats as measurement and remediation platforms for orbital debris

Thanks to their small size and typically shorter development cycles and lower costs, CubeSats have been used as on-orbit technology demonstration missions. A new area of research interest is in situ measurement of the orbital debris environment, with CubeSats as the base of operation. Three ways of performing these measurements have been proposed (the first two have previous flight heritage, with the NASA Space Debris Sensor and the US Air Force Academy FALCON-ODE mission). The first method is direct measurement of impacts on a surface of known area using changes in resistance in a grid or other electrical measurements and using acoustic sensors to detect impacts.

The second method uses a CubeSat as a deployer for calibration targets for ground-based sensors. Since the 1960s spacefaring nations have placed spheres in orbit to use as objects of known size and electromagnetic and optical properties for calibrating

their assets. These calibration objects have typically been large, ranging from 30 cm to 1 m in diameter. The use of radar and optical assets to survey and track orbital debris, which can be smaller than 1 mm in size, requires the ability to calibrate against smaller objects. Deploying small objects with well-known characteristics from a known initial orbit allows ground-based assets to observe calibration targets that are much closer to the sizes of greatest interest to orbital debris modelers. Finally, CubeSats might be able to detect orbital debris (or meteoroids) passing within a certain distance by using high-power pulsed lasers spread out in a sheet perpendicular to the velocity [9]. This detection method allows for direct observation of orbital debris without interacting with them physically, and if two or more laser frequencies are used, operators may be able to actually determine the material composition of the debris object by comparing spectral responses to laboratory measurements.

CubeSats can also perform active debris removal (ADR) missions. These ADR missions usually entail a CubeSat performing a rendezvous operation with a debris object (whether it is an intact rocket body or spacecraft or some other piece of debris), attaching itself to the debris, and then performing some kind of orbit lifetime-reducing maneuver, such as those described in [Section 2.2](#). The RemoveDEBRIS satellite (having mass 100 kg and approximately a 6-cm cube) was developed by a team at the Surrey Space Center and manufactured by Surrey Satellite Technology Ltd. (University of Surrey) [10] and was successfully launched and deployed in early 2019. This mission conducted three experiments, one of which involved capturing a CubeSat using a net, to demonstrate a new technique for grabbing hold of tumbling debris objects. Future missions may use even smaller spacecraft (such as 6U or 12U CubeSats) as the parent object, which could attach drag sails to spent rocket bodies to mitigate their effect on the on-orbit debris environment.

4 Summary

All operators have a responsibility to ensure the sustainability of the on-orbit debris environment, from CubeSats and other small satellites to the massive communications spacecraft in GEO, resulting in every mission minimizing the production of orbital debris during normal operations. Certain launch providers and licensing authorities have requirements that must be fulfilled before receiving permission to launch and operate a CubeSat. These requirements generally deal with on-orbit (production of orbital debris from explosions or collisions) and ground safety (casualty risk to people on the ground from reentering satellites and orbital debris). While CubeSats may not pose a large threat to the orbital debris environment themselves, they may interact with the rest of the on-orbit population (both of active satellites and debris). Ensuring that the on-orbit lifetime is <25 years after the end of the mission (and ideally, no longer than twice the operational lifetime) is a step toward sustainability that any operator can take.

CubeSat operators have a special opportunity: the shorter design-to-launch cycle may allow for innovative uses of space that otherwise would be impossible for a larger, more expensive mission. CubeSats not only can be operated in a way that

minimizes their effect on the debris environment but also can actually serve to improve our understanding of the size, shape, number, and composition of the objects in orbit.

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